

How Transfer Learning Can Help Remote Far-Field EM Side-Channel Attacks

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Abstract. Recently exploited Far-Field EM Side-Channel Attacks (FEM-SCAs), also called ‘screaming channel’ attacks, have emerged as a significant threat to the widely deployed IoT edge devices. By analyzing EM emissions from mixed-signal chips’ RF modules, adversaries can extract secrets from chips over long distances. Existing works conduct remote attacks using deep-learning models trained on ‘clean’ EM traces captured via coaxial cables, while overlooking the fact that long-distance traces suffer from noise and interference. In this paper, we investigate to what extent the Transfer Learning (TL) scheme can help the FEM-SCA model to obtain a better generalization capacity. We experimentally show that transferring a model trained on ‘clean’ traces to profiling traces captured in the victim’s environment reduces the required traces by 12.6% at a 15-meter distance for AES subkey recovery, compared to the baseline model. In addition, we analyze and compare different TL strategies to provide a comprehensive understanding of their impact on FEM-SCAs.

Keywords: Far field EM emission · side-channel attack · Bluetooth · transfer learning · AES.

1 Introduction

The widespread deployment of IoT edge devices has already made them attractive victims of malicious activities. Among various threats, Side-Channel Attacks (SCAs) have emerged as an effective way to compromise physical implementations of cryptographic algorithms. Unlike traditional cryptanalysis, SCAs aim to bypass the inherent theoretical strength of algorithm designs and extract secret keys from their implementations by exploiting unintentional, non-principal physical leakage during device operation. In addition to compromising cryptographic implementations [30], SCAs have also been successfully adopted to reverse deep-learning model architectures [14, 2], monitor user-browser information [44, 45], steal keystroke-based passwords [22, 11], and track code execution [18, 28]. However, most commonly exploited SCA techniques require physical proximity to the victim, which limits attack practicality in numerous scenarios. In actual attack

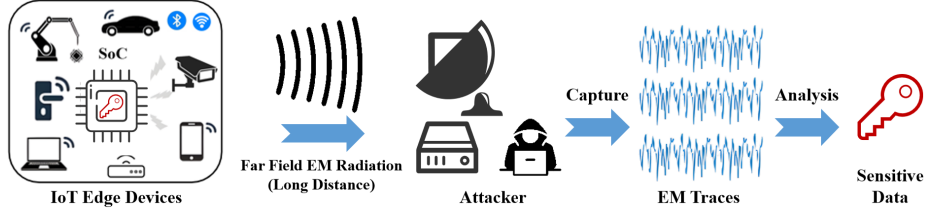


Fig. 1: An overview of the FEM-SCA framework.

scenarios, it is difficult for adversaries to gain access to capture side-channel traces generated by victim devices for further analysis.

Recently, Far-Field EM Side-Channel Attacks (FEM-SCAs) or screaming channels [9], are proposed, enabling key extraction from Bluetooth devices at distances up to 15 m. By exploiting coupling between CPU EM emissions and RF carrier signals from on-chip circuitry, FEM-SCAs enable remote attacks without physical access (see Fig. 1), posing a serious threat to IoT edge devices. In [9], AES key recovery from far-field EM leakage is first demonstrated using averaged measurements. Template attacks extends the attack distance to 15 m, but still rely on extensive measurements and key enumeration [8]. Later, [39] introduces Convolutional Neural Network (CNN)-based deep learning into FEM-SCAs, enabling key recovery from remotely captured traces. However, the limited discriminative information in far-field signals constrains feature learning. To address this issue, [40, 34] train models on ‘clean’ traces collected via coaxial cables with controlled noise augmentation, significantly improving attack performance and reducing the number of required traces. Camurati et al. [7] further establishes the theoretical basis of far-field EM emissions as a side channel. In 2026, [15] exploits frequency diversity to aggregate multi-frequency leakage, improving robustness and enabling successful attacks up to 30 m in RF-polluted environments.

The primary challenge is that long-distance traces are often contaminated by noise and interference, making them difficult for models trained on ‘clean’ traces to recognize and reducing attack efficiency. Moreover, existing FEM-SCAs [9, 8, 39, 40, 19, 20, 1] are mainly evaluated in controlled environments such as anechoic chambers or office corridors, without considering realistic interference sources. In this paper, we leverage transfer learning (TL) to improve model generalization under noisy conditions by transferring knowledge from high-SNR traces, thereby bridging the gap between controlled experimental settings and practical attack scenarios. Our contributions are summarized as follows.

1. We propose a novel transfer learning based FEM-SCA scheme designed to collaboratively leverage cable-captured ‘clean’ traces and remotely captured ‘noisy’ traces. This approach aims to achieve an optimal balance between feature extraction and environmental generalization.

2. We experimentally show the proposed CNN model, trained on ‘clean’ traces and transferred to ‘noisy’ traces, is feasible to recover the AES key by using only 298 traces at 15 m, which is 12.6% more efficient than the SOTA result.

3. We further evaluate different TL strategies to provide a comprehensive understanding of their impact on FEM-SCAs.

The paper is organized as follows. In Section 2, we review the related knowledge about FEM-SCAs, DLSCAs, and the TF scheme. Section 3 introduces the proposed framework. Section 4 presents the experimental setup. Section 5 describes the dataset. Section 6 shows the attack results. Finally, Section 7 concludes the paper.

2 Background

In this section, we first present the background of far-field EM side-channel leakage. We then review the current advances in side-channel attacks based on deep-learning techniques. Next, we introduce the concept of transfer learning.

2.1 Far-Field EM Side Channel

Mixed-signal circuits integrate digital and analog modules on a single silicon chip, where clock switching in the CPU core generates square-wave noise that can affect the sensitive RF block. We use f_s to denote the clock frequency of the square wave. The clock square wave can be decomposed into a set of harmonic components. The amplitude of each harmonic is determined by Fourier-series coefficients A_n , which are governed by the duty cycle τ of the square-wave signal. Due to non-ideal characteristics of practical clock signals, even-order harmonic coefficients are non-zero. Formula (1) gives the Fourier transform $S(f)$ of the clock square wave.

$$S(f) = \sum_{n=-\infty}^{+\infty} A_n \delta(f - nf_s) \quad (1)$$

The side-channel leakage generated during software execution may be modulated by the clock square wave. Specifically, the crypto block $c(t)$ is modulated by the clock square wave $s(t)$, yielding $c_1(t) = c(t) \cdot s(t)$ in time domain. Its frequency-domain representation is given below.

$$C_1(f) = C(f) * S(f) = \sum_{n=-\infty}^{+\infty} A_n C(f - nf_s) \quad (2)$$

This modulated signal carrying side-channel information can inadvertently mix with the baseband signal of the Voltage-Controlled Oscillator (VCO) in the analog circuit, a process facilitated by substrate coupling [5]. The RF block and VCO then further modulate the signal to a high frequency according to the wireless transmission protocol. The radio carrier is denoted by $e^{i2\pi f_c t}$, where f_c is the carrier frequency. The resulting signal in the time and frequency domains

is given by (3), and is subsequently transmitted through the on-chip antenna.

$$\begin{aligned} c_2(t) &= \sum_{n=-\infty}^{+\infty} A_n c(t) e^{i2\pi(nf_s + f_c)t} \\ C_2(f) &= \sum_{n=-\infty}^{+\infty} A_n C(f - nf_s - f_c). \end{aligned} \quad (3)$$

On the receiver side, if the center receiving frequency is set to $Nf_s + f_c$, the received signal is given by formula (4). To recover the side-channel signal $c(t)$, the adversary can use a filter.

$$\begin{aligned} r(t) &= \sum_{n \neq N} A_n c(t) e^{i2\pi(n-N)f_s t} + A_N c(t) \\ R(f) &= \sum_{n \neq N} A_n C(f - (n - N)f_s) + A_N C(f). \end{aligned} \quad (4)$$

In addition to targeting unprotected AES implementations as described in Section 1, FEM-SCAs have also been applied to post-quantum cryptographic implementations [37] and Bluetooth re-keying protocols [10].

2.2 Deep Learning Side-Channel Attack

During the past two decades, SCA techniques have evolved from classical methods such as Correlation Power Analysis (CPA) [4] to more advanced approaches, including Template Attacks (TAs) [12] and deep learning-based approaches. Deep Learning (DL) techniques have become extremely popular for their exceptional ability to identify complex patterns and make accurate classifications from extensive datasets. Generally, a well-trained deep-learning model can make the attack several orders of magnitude more efficient than conventional signal processing methods [13, 16, 31]. Furthermore, various neural networks, including CNNs, are increasingly being utilized by adversaries to overcome countermeasures against SCAs [6, 24]. In [6, 35], the CNN-based strategy markedly simplifies the attack process, eliminating the need for trace alignment and the identification of Points of Interest (PoI). Afterwards, [3] investigates the impact of various hyperparameters of the neural network on DLSCA, introducing the widely recognized benchmark, the ASCAD dataset. Next, the Denoising Autoencoder (DAE) is applied in [42] to degeneralize the noise and countermeasure effect before the training process. Following this direction, Hu et al. [21] further design a multi-loss DAE model which makes the power and near-field EM based SCAs more efficient. In 2022, [37] presents the first deep learning-based FEM-SCA on the Saber Key Encapsulation Mechanism (KEM), a finalist in the NIST post-quantum cryptography standardization project. In [38], it is demonstrated that their deep-learning model could achieve a 90% probability of recovering a bit of the session key. Next, their subsequent work employs deep neural networks to compromise FPGA implementations of masked Kyber-512, as demonstrated in [23]. In 2026, [17] proposes PD-Net, a domain-generalization framework that learns device-invariant

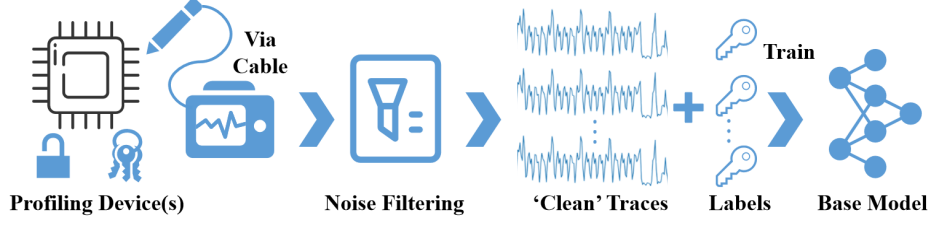


Fig. 2: An overview of the base stage in the proposed TF-based FEM-SCAs.

leakage representations via feature disentanglement and distribution alignment, enabling zero-shot attacks across heterogeneous devices and leakage modalities without target-device data. Similarly, [26] studies the transferability of profiling attacks across implementations and devices, highlighting robust leakage representations and model portability. In addition, [25] introduces differentiable search into non-profiled side-channel attacks, where gradient-based optimization guides key recovery and improves secret key extraction efficiency.

Except for certain exceptions [33, 41], DL techniques are predominantly employed in profiled SCAs, which typically involve two stages: profiling and attack.

During the profiling stage, adversaries are presumed to have full control over one or more profiling device(s), identical to the victim’s and running the same cryptographic algorithm. This allows the attacker to collect numerous side-channel traces and associated information. A chosen model is trained to establish a leakage profile linking secret-dependent values to traces.

During the attack stage, it is common in the community to assume that adversaries could obtain a small amount of side-channel traces and use the pre-trained model to deduce the secret. In FEM-SCAs, the requirement for physical access is waived as the attacker can obtain traces remotely.

2.3 Transfer Learning

Transfer learning is a machine learning technique in which a model developed for one task is reused as the starting point for a model on a related task. Instead of training a model from scratch, transfer learning leverages knowledge from previously trained models to improve learning efficiency and generalization in new but similar tasks. In the context of SCA, transfer learning has been employed to enhance the performance of models trained to extract sensitive information from cryptographic implementations. Typically, SCAs require profiling traces from a similar device in a controlled environment. However, the distribution of these traces often differs significantly from that in real-world attack scenarios due to environmental noise, device variations, or different operating conditions. For these tasks, transfer learning could be a good solution. For example, [46] proposed a DLSCA framework based on transfer learning to adapt the DL classifiers from the profiling devices to a new target device. In FEM-SCAs, two existing works [39, 40] have revealed the performance gap between

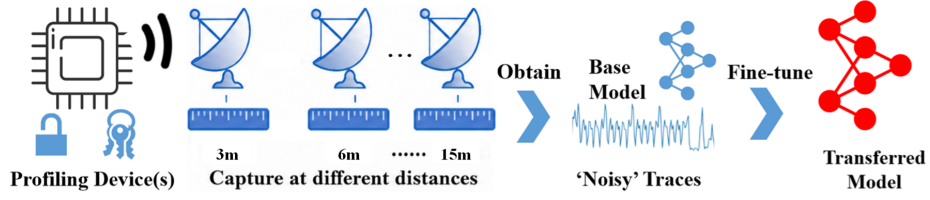


Fig. 3: An overview of the transfer stage in the proposed TF-based FEM-SCAs.

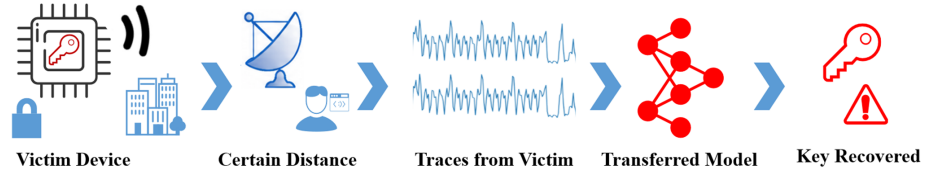


Fig. 4: An overview of the attack stage in the proposed TF-based FEM-SCAs.

‘clean’ near-field traces and noisy long-distance measurements. In [39], the CNN model is trained on remote-captured ‘noisy’ traces to match both the leakage feature and the environmental conditions. However, because traces captured at long distances cannot provide enough leakage features, the model does not fit the patterns fully and requires 10K traces for the 15-m attack. In [40], the CNN model with the same architecture is trained on ‘clean’ traces captured by using the coaxial cable directly touched to the RF block of the chip. These traces provide high-level leakage features, and the models learn the correlation between key-dependent values and traces. Consequently, the number of traces required for the same 15-m attack drops to 340 even though the model is not trained to fit the environmental conditions. In this case, transfer learning might be a good way to bridge the domain gap between the controlled profiling settings (‘clean’ traces) and the real-world environment (‘noisy’ traces). Using the knowledge gained from high-quality profiling traces, models can adapt to noisy and diverse attack traces with reduced data requirements and improved attack accuracy.

3 Transfer Learning-based FEM-SCA

In the proposed transfer learning-based FEM-SCA framework, the methodology is systematically refined into three distinct stages: base, transfer, and attack. In the base stage, a neural network is trained on clean traces to learn leakage-dependent features. In the transfer stage, the learned knowledge is adapted to real-world attack conditions. Finally, the trained model is applied to traces collected from the target device to recover the secret key.

3.1 Base stage

This section describes how we train a neural network as the base model from the very beginning. Fig. 2 illustrates how the base model is prepared in the proposed framework. Let $\mathcal{T} = \{\mathcal{T}_1, \dots, \mathcal{T}_X\}$, where $\mathcal{T}_i \in \mathbb{R}^m$, for $i \in \{1, \dots, X\}$, be a set of X traces representing the computation of the last round of AES which are captured from the profiling device(s) for randomly generated plaintext $\mathcal{P}_i \in \{0, 1\}^{128}$ and a fixed key $K \in \{0, 1\}^{128}$. In the Base stage, traces in $\mathcal{T}$ are captured by using the coaxial cable directly touched to the on-chip antenna so that the EM emissions are recorded without radiating out. Besides, each trace in $\mathcal{T}$ is the average of 100 measurements with the same encryption (the same key and plaintext) to mitigate the effect caused by the noise of the circuit. By doing this, we aim to generate ‘clean’ traces to represent the execution of AES with strong key-dependent feature representation.

Let $\mathcal{C}_i \in \{0, 1\}^{128}$ be the ciphertext which is generated when the trace $\mathcal{T}_i$ is captured. Let $\mathcal{C}_{i,k}$ denote the k th byte of the ciphertext $\mathcal{C}_i$, $k \in \{0, 1, \dots, 15\}$.

Since we use the divide-and-conquer strategy, we train a separate network for each subkey K_k , $k \in \{0, 1, \dots, 15\}$. In this paper, we focus on the first subkey and the process of recovering other subkeys is the same.

To train a network for classifying a subkey K_k , each trace $\mathcal{T}_i \in \mathcal{T}$ is assigned a label, $l_k(\mathcal{T}_i)$, computed as

$$l_k(\mathcal{T}_i) = \mathcal{C}_{i,k} \oplus RK10_k, \quad (5)$$

where $RK10_k$ is the k th byte of the last round key $RK10$. It is easy to find out that the value of $l_k(\mathcal{T}_i)$ is equal to the value of the S-box output in the last round when the k th byte of $\mathcal{C}_i$ is calculated.

The resulting labeled set of traces is used to train the base neural network $\mathcal{N}_k : \mathbb{R}^m \rightarrow \mathbb{I}^{256}$, which maps a trace $\mathcal{T}_i \in \mathbb{R}^m$ into a *score* vector $S_{i,k} = \mathcal{N}_k(\mathcal{T}_i) \in \mathbb{I}^{256}$ whose elements $s_{i,k,j}$ represent the probability that the S-box output in the last round is equal to $j \in \{0, 1, \dots, 255\}$ when the k th byte of $\mathcal{C}_i$ is computed:

$$s_{i,k,j} = \Pr(\mathcal{C}_{i,k} \oplus RK10_k = j). \quad (6)$$

3.2 Transfer stage

After being sufficiently trained on ‘clean’ traces, the base model $\mathcal{N}_k$ is further fine-tuned on remotely captured traces to improve generalization. Fig. 3 illustrates how the base model is fine-tuned during the transfer stage.

Specifically, traces are initially captured at distances of $\{3, 6, 9, 12, 15\}$ meters from the profiling devices to form a transfer set, denoted as $\mathcal{T}' = \{\mathcal{T}'_1, \dots, \mathcal{T}'_Y\}$, where $\mathcal{T}'_i \in \mathbb{R}^m$, for $i \in \{1, \dots, Y\}$. At this stage, each trace is retained in its raw form without undergoing any averaging process. Every trace in $\mathcal{T}'$ is labeled using the same strategy as described previously. The base model $\mathcal{N}_k$ is further fine-tuned on $\mathcal{T}'$ with selected layers frozen (i.e., rendered non-trainable), preserving key-dependent features learned from clean traces while adapting to

far-field conditions. This transfer stage bridges the gap between ideal profiling conditions and real-world attacks. The key idea is to balance preserving general key-dependent features learned from high-quality profiling data with adapting to distorted far-field signals, preventing catastrophic forgetting while enabling specialization to the new data distribution.

In the following sections, we call the model after this process as the transferred model.

3.3 Attack stage

This section describes how we use the fine-tuned neural networks to obtain the secret key from the victim device (see Fig. 4).

Let $\hat{\mathcal{T}} = \{\hat{\mathcal{T}}_1, \dots, \hat{\mathcal{T}}_Z\}$, where $\mathcal{T}_i \in \mathbb{R}^m$, for $i \in \{1, \dots, Z\}$, be a set of Z traces captured from the device under attack. In this case, each trace is captured at 15 m distance to the victim device. At this stage, each trace is retained in its raw form without undergoing any averaging process. Next, the trained network $\mathcal{N}_k$ is used to recover the subkeys K_k from $\hat{\mathcal{T}}$ for every $k \in \{0, 1, \dots, 15\}$ as follows.

The traces $\mathcal{T}_i \in \hat{\mathcal{T}}$ are applied to $\mathcal{N}_k^e$ in order and the most likely label $\tilde{l}_k$ for $\mathcal{T}_i$ is determined among all candidate labels as

$$\tilde{l}_k = \arg \max_{j \in \{0, 1, \dots, 255\}} \left(\prod_{p=1}^i s_{p,k,j} \right), \quad (7)$$

where $s_{p,k,j}$ is the j th element of the score vector $S_{p,k} = \mathcal{N}_k(\mathcal{T}_p)$ of a trace $\mathcal{T}_p \in \hat{\mathcal{T}}$ which precedes $\mathcal{T}_i$ in $\hat{\mathcal{T}}$. Once $\tilde{l}_k = l_k(\mathcal{T}_i)$, the classification is successful.

To verify if $\tilde{l}_k = l_k(\mathcal{T}_i)$, we permute $\hat{\mathcal{T}}$ r times to get the test sets $\hat{\mathcal{T}}_1, \dots, \hat{\mathcal{T}}_r$ and apply them to $\mathcal{N}_k^e$. The point where the rank of K_k for $\mathcal{N}_k^e$ tested on $\hat{\mathcal{T}}_j$, $R_j^e(K_k)$, reaches 0 in the majority of test sets $\hat{\mathcal{T}}_1, \dots, \hat{\mathcal{T}}_r$ is the end point.

3.4 Evaluation Metric

Within the side-channel community, Guessing Entropy (GE) [32] is the most widely used evaluation metric for assessing how effectively an adversary can exploit side-channel leakage to recover cryptographic keys. GE quantifies the uncertainty of the secret key given a specific number of traces, with higher values indicating lower attack success probability.

When recovering 8-bit subkeys individually, Partial Guessing Entropy (PGE) is used instead of GE [29]. PGE represents the expected *rank* of the correct subkey k^* in the key guessing vector g , generated by the model. After classifying a set of traces, g encodes the probabilities of all possible subkey values. If the correct subkey k^* has the highest probability (appearing first in $\text{Sort}(g)$), its rank is 0. The PGE is computed as the average rank of k^* over multiple evaluations.

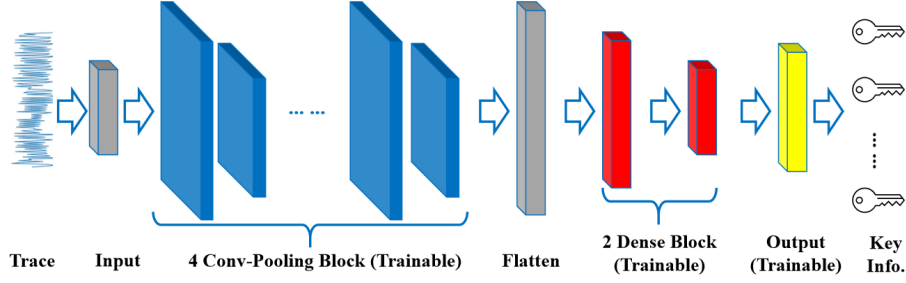


Fig. 5: An illustration of the SOTA CNN model in [40] for FEM-SCA against nRF52 implementation of AES.

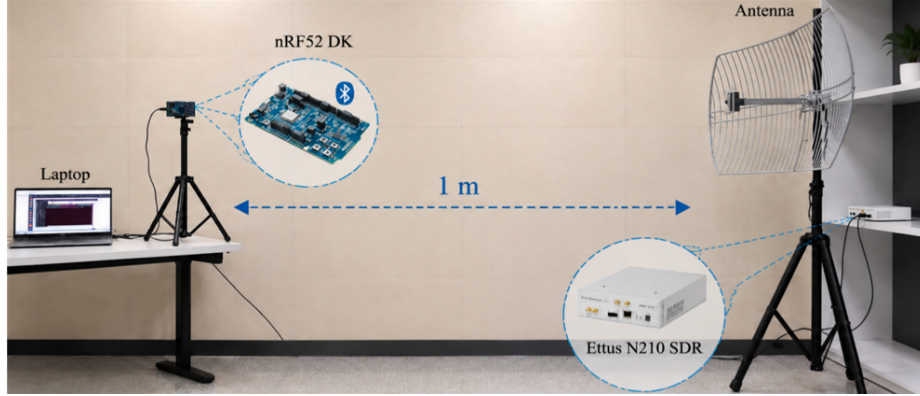


Fig. 6: Experimental setup for far-field EM trace acquisition from the victim nRF52 DK using an Ettus N210 SDR and a parabolic antenna, taking the 1-m setting as an illustration.

4 Experimental Setup

The target device is a Nordic Semiconductor nRF52832 System-on-Chip (SoC) Development Kit (DK), compatible with Bluetooth 5 and capable of transmitting data at 2 Mbps. In the context, we call the target DK as nRF52 device. It features an ARM Cortex-M4 processor operating at 64 MHz. The microcontroller employs the C-based *TinyAES* implementation, utilizing a 128-bit key, and performs AES encryption periodically during the capture process. In the following experiments, we use one nRF52 device for profiling and fine-tuning, while the remaining devices serve as victim devices for the 15 m evaluation scenario.

To capture far-field EM side-channel emissions during nRF52 execution, we use an Ettus N210 Software Defined Radio (SDR) as the core component.

At the base stage, a coaxial cable connects the profiling device to the SDR for high-quality signal acquisition. Its shielding suppresses external radio frequency interference (RFI) and minimizes signal attenuation, preserving weak electro-

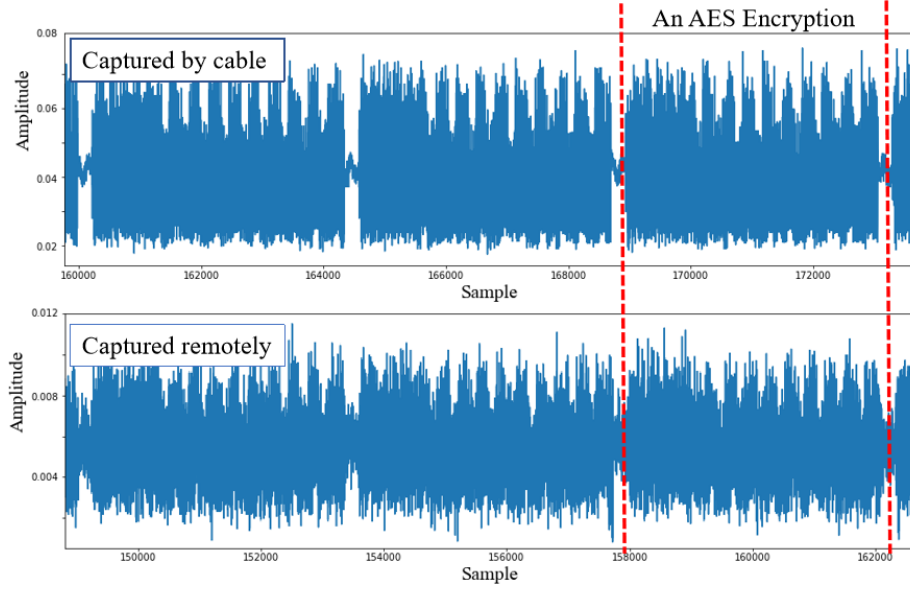


Fig. 7: Traces captured via cable (top) and at a distance of 3 m (bottom). Note that the amplitude of the upper waveform is almost 8-fold larger.

magnetic leakage. Consequently, the SDR captures clean traces for training. We use the training set from [40], whose traces satisfy these requirements.

At the transfer and attack stages, the setup is adapted for far-field signal acquisition. The coaxial cable is replaced by a TL-ANT2424B directional grid parabolic antenna directly connected to the SDR. With a 24 dBi gain, the antenna captures weak EM emissions from the target device over long distances.

For the attack stage, we use the testing set from in [40], which contains sufficient 15-m traces for performance evaluation. For the transfer stage, the traces are collected by ourselves.

For transfer-stage data acquisition, the SDR center frequency f_c is configured as follows.

$$f_c = f_{\text{Bluetooth}} - 2f_{\text{clock}} \quad (8)$$

$f_{\text{Bluetooth}} = 2.4\text{GHz}$ is the Bluetooth channel center frequency and $f_{\text{clock}} = 64\text{MHz}$ is the frequency of the CPU clock. As a result, we can derive that the center frequency f_c of the SDR is set to $f_c = 2.272\text{GHz}$. Notice that this is different from existing FEM-SCA case studies [9, 8, 39, 40, 19, 34], where they normally receive the leakage signal at 2.528 GHz. We adjust the center frequency f_c based on extensive experiments in our measurement environment. We observe strong interference across most harmonic frequencies, except at $f_c = 2.272\text{GHz}$, which is consistent with the setting in [36].

The deep learning model used in the following experiments follows the same architecture as [40] for further comparison. Fig. 5 shows the SOTA CNN model

in [40] for FEM-SCA against the nRF52 AES implementation. To ensure a fair comparison, we use the same optimizer (RMSprop) and loss function (categorical cross-entropy) as in [40].

4.1 Baselines

In our experiments, we compare the proposed transfer learning-based FEM-SCA technique with the following three recent DLSCA advances. All three baseline models are trained on 40K ‘clean’ traces captured from the profiling device via coaxial cable, with 100 repetitions.

1. **Switch-T Transformer** [27]. It is a multi-task Transformer with the Elastic Weight Consolidation (EWC) mechanism to reduce catastrophic forgetting.

2. **LC-CNN** [43]. A CNN model that optimizes Label Correlation (LC) for faster convergence.

3. **SOTA CNN** [40]. This CNN model achieves the state-of-the-art (SOTA) FEM-SCA result at 15 m distance to the victim nRF52 implementation of AES, which is to use 341 traces to recover a 10th-round subkey.

As shown in Fig. 5, the SOTA CNN architecture consists of three components: the convolutional block, dense layers, and the output layer. During transfer learning, we evaluate different fine-tuning strategies by selectively adapting these components, as detailed in the following subsections.

5 Dataset

Firstly, we use the training set in [40] as the base set $\mathcal{T}$, in which the traces are captured from the five profiling nRF52 implementations of AES by using the coaxial cable. For these five profiling devices, we call them PD1 - PD5.

To build $\mathcal{T}$, [40] collects 40K traces from each profiling device, resulting in 200K traces in total, each associated with its corresponding ciphertext and key. Each ‘clean’ trace is obtained by averaging 100 repeated encryptions under identical conditions. The top plot in Fig. 7 shows representative coaxial-cable traces, where three AES executions can be clearly observed.

We construct the fine-tuning set $\mathcal{T}'$ by collecting traces from another nRF52 DK implementation of AES at distances of $\{3, 6, 9, 12, 15\}$ meters. For this device, we denote it as TD0. As illustrated in Fig. 6, the far-field EM traces are captured remotely from TD0 using the SDR. A total of 5K traces are collected, with an equal number of traces from each distance. The bottom plot of Fig. 7 shows representative 3-m traces in $\mathcal{T}'$, where two key differences can be observed compared to shorter distances: (1) significantly attenuated signal amplitude and (2) substantially reduced prominence of the leakage peak.

For the attack stage, we evaluate the model on a publicly available dataset from [40], selected to reflect a realistic attack scenario. The test set consists of far-field EM traces captured at 15 m in a typical office corridor environment. This setting introduces common infrastructural noise, multi-path reflections, and

Table 1: Number of traces required by the base model with learning rate 10^{-5} to recover the AES subkey at 15 m.

Victim Device	VD1	VD2	VD3	VD4	VD5	AVG
# Traces	516	434	687	578	288	500.6

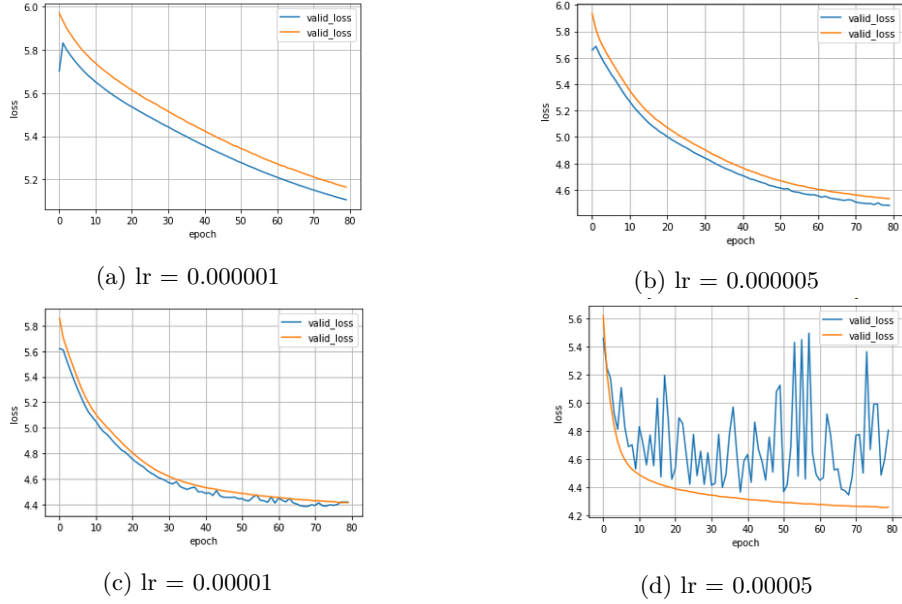


Fig. 8: Training vs. validation loss of the base model with different learning rates.

other unpredictable environmental variables that are absent in controlled laboratory settings, providing a robust test for the model’s generalization capability. The dataset contains 25K traces collected from five nRF52 victim devices, evenly distributed. We denote these devices as VD1 - VD5. This partitioning enables evaluation of model performance not only on a single device but also across multiple instances of the same device type.

6 Experimental Results

In this section, we first demonstrate the testing results of the base model before the transfer stage. Next, we show how different freezing strategies can affect the attacking results of the transferred model.

6.1 Base Model

At the beginning, we apply a grid search approach to find the optimal learning rate to build the base model. The training is carried out for a maximum of

Table 2: Number of traces required by $\mathcal{N}^0$ to recover the subkey at 15 m.

Victim	VD1	VD2	VD3	VD4	VD5	AVG
# Traces	443	192	414	318	125	298.2

Table 3: Number of traces required by $\mathcal{N}^1$ to recover the subkey at 15 m.

Victim	VD1	VD2	VD3	VD4	VD5	AVG
# Traces	532	496	603	452	246	465.8

80 epochs with batch size 256. For each training task, we randomly select 40K traces from the base training set for validation.

We test the base model with the learning rate 10^{-5} on testing traces, which are captured at 15 m distance with 1 repetition. This is close to the real attack scenario, as traces are captured in an office corridor that is not dedicated for the attack experiments. Table 1 shows the test results (PGE results) of the base model on traces captured at 15 m distance to VD1 - VD5.

From Table 1, the base model requires 500.6 traces on average to compromise the AES implementation at 15 m before transfer to noisy traces. This result is close to the first column of Table 6 in [40] under the same setting, confirming that the base model training strategy is well designed.

6.2 Transferred model

In transfer learning, layer freezing preserves knowledge learned from the source domain by preventing selected network layers from being updated during training. It is particularly effective when the source and target datasets share common features, improving efficiency while retaining useful representations.

Our base model consists of three trainable components (see Fig. 5). To identify which components should be fine-tuned on noisy traces for better generalization, we derive four variants from the base architecture as follows.

1. $\mathcal{N}^0$: All layers remain trainable.
2. $\mathcal{N}^1$: Only the convolutional block is non-trainable.
3. $\mathcal{N}^2$: Only the dense layers are non-trainable.
4. $\mathcal{N}^3$: Only the output layer is non-trainable.

All models are trained for up to 100 epochs with a batch size of 256. For each run, 400 traces are randomly selected from the fine-tuning set $\mathcal{T}'$ for validation. We evaluate each model on 15 m traces separately Fig. 9 presents the PGE results of the four models. Each experiment is repeated 100 times with randomly sampled test traces, and the final results are averaged.

Table. 2 - 5 summarize the attack results of model $\mathcal{N}^0$ - $\mathcal{N}^3$ on traces captured at 15 m distance to the victim devices.

Table 4: Number of traces required by $\mathcal{N}^2$ to recover the subkey at 15 m.

Victim	VD1	VD2	VD3	VD4	VD5	AVG
# Traces	548	410	402	512	137	407.0

Table 5: Number of traces required by $\mathcal{N}^3$ to recover the subkey at 15 m.

Victim	VD1	VD2	VD3	VD4	VD5	AVG
# Traces	512	632	852	790	462	649.6

From Table 2 to Table 5, we can find that not all transferred models generalize better than the base model in Table 1. $\mathcal{N}^3$ requires 649.6 traces on average to attack the AES implementation at 15 m, which is less efficient than the base model. Among all models, $\mathcal{N}^0$ achieves the best classification performance at 15 m. only 298.2 traces on average to recover the AES subkey, improving the SOTA result in [40] by 12.6%. This indicates that adapting the entire model to the target attack environment is crucial for maximizing FEM-SCA performance. It further suggests that both low-level (convolutional) and high-level (dense) features capture complementary patterns that contribute to attack success.

Additionally, results of Table 1 and 2 show that the base model needs an average of 500.6 traces before transfer-learning fine-tuning. Full-layer fine-tuning over a small set of ‘noisy’ traces boosts attack efficiency by 40.4%. This indicates that fine-tuning the entire model (rather than freezing layers) on noisy data improves generalization, since noisy traces contain discriminative leakage features missed by the pre-trained base model.

From Table 2 to Table 5, the attack efficiency is ranked as: $\mathcal{N}^0$, $\mathcal{N}^2$, $\mathcal{N}^1$, $\mathcal{N}^3$. This indicates that low-level feature extraction in convolutional layers is more critical for FEM-SCA performance than high-level decision-making in dense layers, while excessive layer freezing significantly reduces adaptability. These results suggest that effective transfer learning should preserve the trainability of convolutional layers while selectively freezing higher layers. They further highlight that side-channel leakage is better captured through low-level signal characteristics, emphasizing the importance of feature extraction in far-field SCA.

The superiority of $\mathcal{N}^0$ demonstrates that maximum adaptability is essential for effective transfer learning. Fine-tuning all model parameters on noisy data achieves better generalization than partially frozen architectures, indicating that the distribution shift between training and real-world conditions requires holistic model adaptation.

The critical insight comes from comparing the partially frozen models. $\mathcal{N}^2$ (frozen dense layers) outperforms $\mathcal{N}^1$ (frozen convolutional block), confirming that low-level feature extraction is more critical for FEM-SCA than high-level decision-making. Convolutional layers act as the front line for processing the raw, noisy electromagnetic traces. When frozen, the model relies on features learned

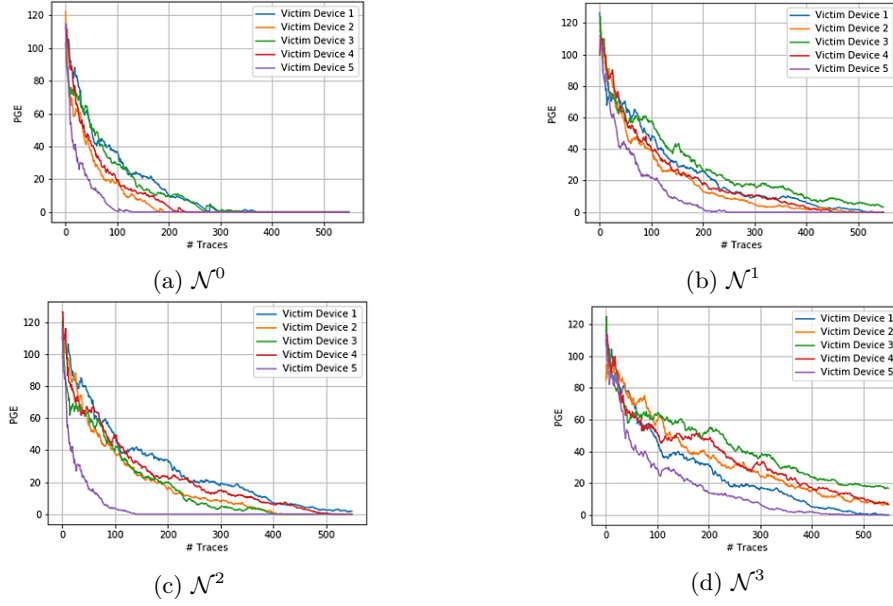


Fig. 9: Average PGE of different models at 15 m (VD1 - VD5). Each curve is averaged over 100 runs.

from a mismatched distribution, which are suboptimal in the new adversarial setting. In contrast, when only dense layers are frozen, the convolutional block can adapt to extract leakage-relevant features from noisy signals, which are then effectively handled by a static classifier.

Furthermore, the poor performance of $\mathcal{N}^3$ (frozen output layer) highlights the importance of output-layer adaptation. Although convolutional layers are essential for feature extraction, the output layer maps learned features to target key predictions and must remain trainable. When frozen, it cannot adapt to the far-field distribution, becoming a bottleneck that degrades classification performance and increases the number of traces required for successful key recovery.

The performance differences stem from feature transferability and task specificity. In side-channel transfer learning, early convolutional layers learn general low-level features (e.g., shapes of traces) transferable across measurement setups, whereas later dense and output layers learn task-specific features and decision boundaries tailored to the original training data.

In the context of FEM-SCA, the remote traces are inherently noisier and contain a different signal profile compared to ‘clean’ traces captured by using the coaxial cable. The $\mathcal{N}^0$ model succeeds because it fine-tunes all specialized layers to the new signal characteristics. In contrast, models that freeze critical layers such as dense blocks or the output layer are forced to map noisy data to a feature space and decision logic designed for a different task, resulting in reduced attack efficiency and more required traces.

Table 6: Average number of traces required by models to recover the target subkey (PGE=0) at 15 m distance (results for 100 tests).

Victim device	D1	D2	D3	D4	D5	AVG
$\mathcal{N}^0$	443	192	414	318	125	298
Switch-T [27]	2409	3063	3918	3358	2134	2976
LC-CNN [43]	1667	1358	3349	>4999	1138	/
SOTA CNN [40]	292	467	256	384	306	341

A successful transfer strategy requires balancing adaptability and stability rather than simply freezing layers. Keeping convolutional layers trainable is essential for capturing leakage-relevant features in new environments, while selectively freezing higher layers can improve stability and reduce overfitting, as shown by $\mathcal{N}^2$. This balance enables adaptation to noisy signals while preserving the benefits of pre-trained knowledge.

6.3 Hypothesis

From the experimental results, we can summarize the following hypothesis.

Hypothesis 1: Convolutional layers capture essential far-field EM leakage patterns, making them critical for attack performance. Consequently, freezing these layers ($\mathcal{N}^1$) significantly degrades efficiency. Unlike computer vision models that learn generic low-level features, side-channel models trained on ‘clean’ traces learn device-specific leakage patterns. Since far-field traces exhibit different frequency characteristics, propagation effects, and signal-to-noise ratios, updating the convolutional layers is necessary to adapt to these changes.

Hypothesis 2: Dense layers primarily refine decision boundaries, so freezing them ($\mathcal{N}^2$) reduces performance but still allows the model to exploit useful features learned by the convolutional layers. Although the dense layers are optimized for the clean data distribution, adapted convolutional features remain sufficient for accurate key classification.

Hypothesis 3: Over-constrained models ($\mathcal{N}^3$) perform poorly because updating only the output layer is insufficient to adapt to new noise conditions. As the internal feature representations change while the output layer remains fixed to the clean-data distribution, a mismatch arises that degrades transfer performance. This result shows that effective transfer learning requires coordinated adaptation across the network.

6.4 Comparison with Baselines

Afterwards, we compare the best $\mathcal{N}^0$ model in the experiments above to the selected baselines in existing DLSCA works. Table 6 reports the average number of traces required to recover the target subkey (PGE=0) at a distance of 15 m. The results show that $\mathcal{N}^0$ exhibits strong performance in terms of attack-efficiency

among the evaluated methods. It yields the lowest average trace requirement across the five victim devices, although performance improvements are uneven for different hardware targets.

In contrast, the other methods show clear limitations. Switch-T requires significantly more traces, making it less efficient. LC-CNN is unstable, failing on device D4 (requiring > 4999 traces) and showing large variations across devices, resulting in an undefined average (denoted by ∞). The SOTA CNN achieves 341 traces on average, outperforming Switch-T and LC-CNN but still below $\mathcal{N}^0$.

Overall, the comparison shows $\mathcal{N}^0$ as the most efficient and reliable model in this benchmark. Its lower trace requirement implies reduced attack cost, faster execution, and higher success probability. The performance gap with other methods, especially the instability of LC-CNN and high trace cost of Switch-T, highlights the effectiveness of the proposed transfer learning strategy for FEM-SCA.

7 Conclusion

In this work, we address a critical limitation of existing FEM-SCA attacks: reliance on deep-learning models trained only on ‘clean’ EM traces, which ignore noise and interference in long-distance captures. By leveraging transfer learning techniques, we show that adapting a pretrained model to target-domain traces significantly improves real-world generalization. Experiments demonstrate a 12.6% reduction in required traces at 15 m for AES implementations.

In addition, our evaluation of transfer learning strategies for FEM-SCAs shows that full network adaptation achieves the highest attack efficiency, while partial freezing highlights the role of convolutional layers in low-level feature extraction. Excessive freezing degrades performance below the baseline, providing guidance for model transfer in far-field side-channel analysis.

Future work includes evaluating model performance under varying noise conditions and exploring optimized training strategies such as selective layer freezing. A key direction is the design of countermeasures against FEM-SCA, particularly given the growing deployment of vulnerable IoT edge devices.

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